

Sarvesh Mishra

Roll No.: MT25CS008 M.Tech Computer Science and Engineering

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Portfolio

PROFESSIONAL SUMMARY

M.Tech Computer Science and Engineering student at NIT Mizoram with hands-on experience in **Artificial Intelligence, Machine Learning, Deep Learning, Data Analysis, Computer Vision, Backend Development, REST APIs, and database-driven applications**. Completed a Summer Research Internship at IIT (ISM) Dhanbad, developing and evaluating deep learning-based solutions for **handwritten cheque field information extraction** using real-world and synthetic data. Experienced in **Python, SQL, feature engineering, model evaluation, API integration, application development, and end-to-end AI solution development**. Built applied machine learning and web applications involving data preprocessing, model training, explainable AI, authentication, databases, and deployment. Strong analytical and problem-solving skills with experience translating technical requirements and data-driven problems into practical solutions.

EDUCATION

M.Tech in Computer Science and Engineering

National Institute of Technology, Mizoram

2027

CGPA: 8.00

B.Tech in Computer Science and Engineering

University of Lucknow

2025

CGPA: 8.15

INTERNSHIP EXPERIENCE

Summer Research Intern – IIT (ISM) Dhanbad

May 2026 – July 2026

Dhanbad, Jharkhand, India | On-site Research Internship (SRIS-2026)

Experience Letter

Project: Deep Learning-Assisted Handwritten Cheque Field Information Extraction Model for Bank Cheques

Supervisor: Dr. Soumen Bag, Associate Professor, Department of CSE, IIT (ISM) Dhanbad

- Developed **Faster R-CNN** object detection models with **ResNet50-FPN** and **ResNet18-FPN** backbones to localize five cheque fields: Date, Name, Amount in Words, Amount in Numbers, and Signature.
 - Prepared and evaluated models using **synthetic data (620 images)** and **real IDRBT data (112 images)**, applying transfer learning with 50% of the backbone frozen and testing on unseen SBI and multi-bank cheques.
 - Achieved **98.40% precision, 94.62% recall, 96.47% F1-score, and 98.00% mAP@50** with the best ResNet18-FPN model on unseen SBI cheques.
 - Used **PyTorch, OpenCV, Docker, and CVAT** for end-to-end computer vision model development, dataset preparation, training, evaluation, and experimentation.
- ### Software Development Intern – Nelumbus Technologies LLP
- Remote, India*
- May 2026 – July 2026**
- Experience Letter
- Executed **end-to-end software development tasks**, converting technical requirements into deliverable application features under senior engineering mentorship.
 - Applied software engineering practices for **feature implementation, systematic debugging, testing, and iterative application enhancement**.
 - Coordinated with senior mentors in a remote development environment to clarify requirements, review implementation progress, and incorporate feedback into application features.

PROJECTS

PredictMedix – Medical Cost Prediction System

GitHub | Live Demo

- Designed an end-to-end **machine learning pipeline** covering data preprocessing, exploratory data analysis, feature engineering, model training, prediction, and evaluation using **XGBoost, Pandas, NumPy, and Scikit-learn**.
- Applied **SHAP-based Explainable AI** to quantify feature contributions to predicted medical insurance charges and generated **5-year cost projections**.
- Built and deployed a **Flask web application** with authentication, MongoDB/local JSON database fallback, and PDF report generation.
- Integrated machine learning outputs into an application workflow to provide practical, data-driven insurance cost predictions and explanations.

Travel Quest – Hotel Booking Web Application

GitHub | Live Demo

- Designed a **Node.js and Express.js backend** with MongoDB, implementing CRUD workflows and **RESTful API routes** for listings, users, reviews, and listing management.
- Implemented application security through **Passport.js authentication, session management, owner/author authorization, Joi-based validation, and centralized error handling**.
- Integrated **Cloudinary/local image storage and Mapbox APIs**, supporting search and filtering across 9 listing categories, price alerts, expense tracking, and similar-listing recommendations.

TECHNICAL SKILLS

Programming: Python, C++, C

Artificial Intelligence & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Scikit-learn, PyTorch, XGBoost, Regression, Random Forest, Gradient Boosting, Transfer Learning, Hyperparameter Tuning, Model Evaluation

Data Analysis: Data Analysis, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Pandas, NumPy

Computer Vision & Document AI: Computer Vision, Faster R-CNN, Object Detection, Image Segmentation, SAM, OpenCV, Document Information Extraction

Application Development & APIs: Node.js, Express.js, Backend Development, REST APIs, API Integration, Authentication, Session Management, Error Handling

Databases & SQL: MySQL, MongoDB, Mongoose, SQL, Database Querying, CRUD

Generative AI & Explainable AI: Prompt Engineering, Explainable AI, SHAP

Testing, Evaluation & Tools: Performance Evaluation, Test Data Preparation, Precision, Recall, F1-score, mAP@50, MAE, RMSE, R^2 , Git, GitHub, Docker, VS Code, CVAT

KEY COURSES

Data Structures and Algorithms (DSA), Operating Systems, Computer Networks, Database Management Systems (DBMS), Object-Oriented Programming (OOP), Pattern Recognition, Machine Learning, Deep Learning, Data Mining, Big Data Analytics

CERTIFICATIONS & TRAINING

Artificial Intelligence

1Stop | 2026

Machine Learning

1Stop | 2026

GenAI Powered Data Analytics Job Simulation

Forage | 2026

Prompt Engineering for GenAI

Internshala Trainings | 2024

EXTRACURRICULAR

Author – “Manushya Hone Ki Yatra”

2026

- Authored and published the **first edition (2026)** of a Hindi book exploring life, relationships, family, society, truth, responsibility, and human behavior through personal experiences and reflections (**ISBN 978-93-5937-336-2**).

Student Coordinator – Institute Innovation Council (IIC), NIT Mizoram

February 2026 – April 2026

- Coordinated Ministry of Education-aligned innovation activities, workshops, structured discussions, and collaborative events to strengthen student engagement.